Science & Tech 13 Aug 2026

Semicon 2.0 and IndiaAI Mission Drive India's Chip-to-AI Build-Out

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Key Points

1. Semicon India 2.0 was approved in July 2026 with a ₹1,27,500 crore outlay, scaling up the ₹76,000 crore Semicon 1.0 programme across six pillars — chip design, equipment and materials, fabrication, advanced packaging, R&D and talent.
2. Twelve semiconductor projects have been approved across six states with investment commitments above ₹1.64 lakh crore; three facilities have already begun commercial production.
3. The IndiaAI Mission, with an outlay of about ₹10,372 crore, had expanded shared compute to more than 45,000 GPUs as of June 2026, and 237 projects had used 93.18 lakh subsidised GPU hours by August 2026.

4. Twenty indigenous foundation model proposals — 12 large multimodal models and 8 small language models — were selected from 506 applications; AI Kosh hosted over 14,000 datasets and 331 AI models as of July 2026.
5. India was ranked third globally in the Stanford Global AI Vibrancy Report 2025 and joined the Pax Silica coalition on resilient silicon supply chains at the India AI Impact Summit 2026.

Quick Facts

- Semicon 1.0 outlay: ₹76,000 crore; Semicon 2.0 outlay: ₹1,27,500 crore, approved July 2026
- 12 semiconductor projects approved across 6 states; commitments over ₹1.64 lakh crore; 3 units in commercial production
- Chips to Startup (C2S): EDA tools deployed in 320 academic institutions, over 68,000 students trained
- Design Linked Incentive: 24 semiconductor design projects and 105 startups/MSMEs supported with EDA tools
- 211 chips taped out by 75 institutions by April 2026; 7 fabricated, including at the 12 nm node
- IndiaAI Mission outlay about ₹10,372 crore; 45,000+ GPUs of shared compute as of June 2026
- 58 AI Centres of Excellence approved as of August 2026; 22 already initiated across 13 States/UTs
- 62 AI prototypes developed and 20 AI solutions deployed in public-sector institutions as of August 2026
- 13 Responsible AI projects approved under the Safe & Trusted AI pillar as of July 2026
- Data centre installed capacity grew from 375 MW in 2020 to about 1,575 MW
- India AI Impact Summit 2026 (February 2026): delegations from over 100 countries, declaration adopted by 92 countries and organisations

Schemes & Policies

India Semiconductor Mission (ISM) / Semicon India Programme

Build a domestic semiconductor and display manufacturing and design ecosystem, and act as the nodal agency implementing the semiconductor schemes

Key: Launched in 2021 under the Ministry of Electronics and Information Technology with a ₹76,000 crore outlay; set up as an independent business division within the Digital India Corporation with administrative and financial autonomy. It covers silicon fabs, display fabs, compound semiconductor/silicon photonics/sensor fabs, packaging (ATMP/OSAT) and design.

Semicon India 2.0

Move from building an ecosystem to global ecosystem leadership

Key: ₹1,27,500 crore outlay approved in July 2026, organised around six pillars: chip design, semiconductor equipment and materials, fabrication facilities, advanced packaging, research and development, and talent development. It also adds support for semiconductor IP, chip and system designs.

Design Linked Incentive (DLI) Scheme

Support domestic companies, startups and MSMEs across semiconductor design — integrated circuits, chipsets, systems-on-chip, systems and IP cores

Key: Launched in December 2021 and notified in the Gazette on 21 December 2021; C-DAC is the nodal implementing agency. It offers financial incentives, access to advanced design tools and prototyping support. As per this release, 24 design projects and 105 startups/MSMEs have been supported.

Chips to Startup (C2S) Programme

Build the chip-design workforce by putting industry-grade design tools inside academic institutions

Key: Electronic Design Automation tools deployed in 320 academic institutions with over 68,000 students trained; 211 chips taped out by 75 institutions by April 2026, of which 7 were fabricated at nodes including 12 nm.

IndiaAI Mission

Build a sovereign, inclusive AI ecosystem — compute, datasets, models, applications, skills, startup finance and safe/trusted AI

Key: Approved on 7 March 2024 with an outlay of ₹10,371.92 crore under MeitY, structured around pillars including AI Compute Capacity, Innovation Centre, Datasets Platform, Application Development Initiative, FutureSkills, Startup Financing and Safe & Trusted AI.

AI Kosh

Serve as the national datasets and models platform for AI developers and researchers

Key: The datasets platform pillar of the IndiaAI Mission; hosted over 14,000 datasets and 331 AI models as of July 2026.

National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)

Create research, translation and entrepreneurship capacity in cyber-physical systems

Key: Has established 25 Technology Innovation Hubs at leading academic institutions covering AI, robotics, cybersecurity, quantum technologies and other emerging fields.

National Quantum Mission (NQM)

Seed and scale scientific and industrial R&D in quantum technologies

Key: Approved by the Union Cabinet on 19 April 2023 at a total cost of ₹6,003.65 crore for 2023-24 to 2030-31, implemented by the Department of Science and Technology. Deliverables include quantum computers with 50-1,000 physical qubits in eight years and satellite-based quantum communication over 2,000 km.

Key Data & Numbers

Semicon 1.0 outlay

Semicon 2.0 outlay

₹76,000 crore (2021)

₹1,27,500 crore (approved July 2026)

Semiconductor projects approved

12, across 6 states, over ₹1.64 lakh crore committed

IndiaAI Mission outlay

₹10,371.92 crore (approved 7 March 2024)

Shared GPU capacity

45,000+ GPUs as of June 2026

Subsidised compute used

93.18 lakh GPU hours across 237 projects by August 2026

Foundation models selected

20 of 506 applications — 12 large multimodal models, 8 small language models

AI Centres of Excellence

58 approved; 22 initiated across 13 States/UTs (August 2026)

Data centre capacity

About 1,575 MW, up from 375 MW in 2020

National Quantum Mission outlay

₹6,003.65 crore (2023-24 to 2030-31)

Semicon India 2025 event

350+ exhibitors, participants from 48 countries, 12 MoUs announced

Important Bodies

India Semiconductor Mission (ISM)

Nodal agency for implementing India's semiconductor and display manufacturing and design schemes; an independent business division within the Digital India Corporation with administrative and financial autonomy

Head: Headed by a Chief Executive Officer HQ: New Delhi; parent ministry — Ministry of Electronics and Information Technology

Centre for Development of Advanced Computing (C-DAC)

Premier R&D organisation of MeitY; nodal agency for the Design Linked Incentive Scheme and developer of the PARAM series of supercomputers (beginning with PARAM 8000 in 1991)

Head: Director General

HQ: Pune; established in March 1988 as an autonomous scientific society, after the denial of supercomputer exports to India

AIRAWAT

India's largest AI supercomputing system, installed at C-DAC Pune; it was ranked 75th in the 61st edition of the Top 500 global supercomputing list announced at ISC 2023

Head: Operated by C-DAC HQ: C-DAC, Pune

Department of Science and Technology (DST)

Implements the National Quantum Mission and hosts NM-ICPS Technology Innovation Hubs

Head: Union Minister of Science and Technology; headed administratively by a Secretary

HQ: New Delhi; under the Ministry of Science & Technology

Know the Concept

The semiconductor value chain splits into design (front-end), fabrication, and assembly-test-packaging (back-end)

A fab turns raw silicon wafers into functioning integrated circuits using photolithography at a given process node (12 nm, 28 nm and so on) — it is the most capital-intensive stage. ATMP stands for Assembly, Testing, Marking and Packaging: it takes finished wafers from a fab, cuts and encases the chips, wires them, marks and tests them. OSAT — Outsourced Semiconductor Assembly and Test — is the same work offered as a third-party service rather than done in-house. Display fabrication is a separate line making flat and flexible screens. Design sits before all of this and uses Electronic Design Automation (EDA) software to lay out, simulate and verify a chip before a single wafer is touched; a completed design sent for manufacturing is called a 'tape-out'. India's approved projects span all of these categories, and Micron's Sanand plant in Gujarat — approved by the Union Cabinet on 22 June 2023 with a capital investment of ₹22,516 crore (about USD 2.75 billion) under the Modified ATMP Scheme — is the back-end example.

Simple Analogy: A fab is the printing press that makes the pages; ATMP/OSAT is the bindery that trims, covers and quality-checks the book; design with EDA tools is the manuscript and proofreading that must be finished before anything is printed.

Compare

Aspect	Semicon India 1.0	Semicon India 2.0
Year	Launched 2021	Approved July 2026
Outlay	₹76,000 crore	₹1,27,500 crore
Stated aim	Lay the foundation of a semiconductor ecosystem	Scale the foundation into global ecosystem leadership
Structure	Support across chip design, fabrication, packaging, testing, equipment, materials and	Six pillars: chip design; equipment and materials; fabrication; advanced packaging; R&D; talent

Aspect	Semicon India 1.0	Semicon India 2.0
	talent	development
Design focus	Design Linked Incentive and Chips to Startup support	Adds support for semiconductor IP, chip and system designs

Timeline

March 1988
C-DAC established after supercomputer exports to India were denied; later the nodal agency for the DLI Scheme

2021
India Semiconductor Mission launched with a ₹76,000 crore outlay under MeitY

21 December 2021
Design Linked Incentive Scheme notified in the Gazette of India

19 April 2023
Union Cabinet approves the National Quantum Mission (₹6,003.65 crore, 2023-24 to 2030-31)

22 June 2023
Union Cabinet approves Micron's semiconductor assembly and test facility at Sanand, Gujarat (₹22,516 crore)

7 March 2024

IndiaAI Mission approved with an outlay of ₹10,371.92 crore

April 2026

211 chips taped out by 75 institutions under the design programmes; 7 fabricated

February 2026

India AI Impact Summit 2026 held; declaration adopted by 92 countries and organisations; India joins the Pax Silica coalition

July 2026

Semicon India 2.0 approved with a ₹1,27,500 crore outlay

Global Context

The release places India's position through three markers, all of them from official and published sources. The Stanford Global AI Vibrancy Report 2025 ranked India third globally on AI competitiveness and ecosystem vibrancy, and in 2025 India was the second-largest contributor to GitHub AI projects. On the manufacturing side, India's approach is partnership-led rather than autarkic: the release names technology collaboration with the United States, Japan, Singapore, the Netherlands, Germany and the European Union, and Semicon India 2025 drew exhibitors and participants from 48 countries with 12 MoUs announced. At the India AI Impact Summit 2026 in February 2026, delegations from more than 100 countries and 20 international organisations took part and 92 countries and organisations adopted the declaration. India also joined Pax Silica, a coalition with the United States and other

partners aimed at resilient global silicon supply chains — supply-chain resilience, rather than cost, is the organising idea of the current phase.

Connections

National Quantum Mission

Both NQM (DST, ₹6,003.65 crore, approved 19 April 2023) and the semiconductor programme rest on the same bottleneck — advanced fabrication and specialised processors. NM-ICPS Technology Innovation Hubs cover quantum technologies alongside AI and robotics.

Critical minerals and supply chains

Chip fabrication and advanced packaging depend on materials whose supply is geopolitically concentrated, which is why 'equipment and materials' is a standalone pillar of Semicon 2.0 and why India joined the Pax Silica coalition.

Data centres and energy demand

Data centre capacity rose from 375 MW in 2020 to about 1,575 MW, and AI compute is energy-intensive — linking this topic directly to India's power capacity expansion and non-fossil energy targets.

Digital India Corporation

ISM is not a separate statutory body but an independent business division inside the Digital India Corporation — a favourite distinction in questions about how the mission is actually constituted.

Exam Spotlight

Exam Relevance

UPSC
GS Paper 3 > Science and Technology > Indigenisation of Technology and Developing New Technology
●●●●●

SSC
General Awareness > Government Schemes and Missions
●●●●○

BANKING
General Awareness > Economy and Industrial Policy
●●●○○

Expected Questions

- ★ UPSC may ask a multi-statement question distinguishing the nodal agency, parent ministry and legal form of the India Semiconductor Mission from those of the IndiaAI Mission.
- ★ A pairing question matching missions to implementing departments — IndiaAI Mission to MeitY, National Quantum Mission to DST, DLI Scheme to C-DAC.
- ★ SSC/Railway may ask the outlay figures directly: ₹76,000 crore, ₹1,27,500 crore, ₹10,371.92 crore, ₹6,003.65 crore.
- ★ A conceptual question on the difference between a fab, an ATMP/OSAT unit and a design house.

Topic Frequency

Semiconductors and AI missions have appeared repeatedly in Prelims and Mains since 2022 and are near-certain recurring themes

Key Terms

Tape-out

EDA (Electronic Design Automation)

The point at which a finished chip design is sent for manufacturing; 211 tape-outs by 75 institutions had been recorded by April 2026.

Software used to design, simulate, verify and optimise chips before manufacture, catching errors and cutting design time and cost.

ATMP

Assembly, Testing, Marking and Packaging — the back-end stage that turns fabricated wafers into usable, tested chips.

OSAT

Outsourced Semiconductor Assembly and Test — ATMP work provided as a third-party service to chipmakers.

GPU

Graphics Processing Unit, a specialised processor for graphics and data-intensive computing; the unit in which IndiaAI's shared compute is measured.

Process node

The manufacturing generation of a chip, expressed in nanometres — India has fabricated chips at nodes including 12 nm.

Must Remember

- India Semiconductor Mission — launched 2021, ₹76,000 crore, under MeitY, an independent business division within Digital India Corporation
- Semicon India 2.0 — ₹1,27,500 crore approved July 2026, six pillars
- IndiaAI Mission — approved 7 March 2024, outlay ₹10,371.92 crore, MeitY
- Design Linked Incentive Scheme — notified 21 December 2021, C-DAC is the nodal agency
- National Quantum Mission — approved 19 April 2023, ₹6,003.65 crore, Department of Science and Technology
- C-DAC — established March 1988, headquartered at Pune; AIRAWAT installed there ranked 75th in the Top 500 list at ISC 2023
- Micron's Sanand ATMP plant — approved by the Cabinet on 22 June 2023, ₹22,516 crore (about USD 2.75 billion)

Exam Tips

- The trap here is treating ISM and IndiaAI as one programme. They share a ministry (MeitY) but differ in year, outlay and instrument — ISM 2021/₹76,000 crore for manufacturing, IndiaAI 2024/₹10,371.92 crore for compute and models.

- Do not confuse the DLI Scheme (design, nodal agency C-DAC) with the Modified ATMP Scheme (back-end manufacturing, under which Micron's Sanand plant was approved).

Practice Quiz

Question 1 of 4 Score: 0/0

Factual

The Chips to Startup (C2S) programme, under which Electronic Design Automation tools were deployed in 320 academic institutions, forms part of which initiative?

A India Semiconductor Mission

B IndiaAI Mission

C National Quantum Mission

D National Mission on Interdisciplinary Cyber-Physical Systems

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